



Measuring management accounting practices using textual analysis

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ABSTRACT

Management accounting is an important component of managerial decision-making to help guide the overall business strategy in organisations. Appropriate measures of management accounting practices (MAPs) allow academics and practitioners to assess the organisations' MAP adoption effectively. In this study, based on qualitative information disclosed in annual reports from 2003 to 2019, we construct and validate novel and comprehensive measures of firm-level MAPs using the word embedding model Word2Vec. We illustrate the predictive validity of our measures by examining the determinants and value creation function of fitted MAPs. Our novel measures of MAPs not only provide a solid foundation for large scale sample empirical analyses of management accounting research but also pave a new way to investigate MAPs as a package.

Companies with advanced management often apply the standard cost, comprehensive budget management, performance evaluation, cost analysis, and so forth, and gradually build the basis for data analysis. And differences analysis can further help management to gradually understand the nature of the business and make better judgments accordingly; thus, companies will achieve value creation in the end.

Eric Yang, Bosch's (China) vice CFO

1. Introduction

The design of management accounting practices (MAPs) aims to gain continuous improvement and ultimately achieve organisational objectives (Albright and Lam, 2006). Although prior studies have provided extensive evidence regarding MAP adoption (Ax and Greve, 2017; Dalla Via et al., 2018; Hiebl and Richter, 2018), empirical research on MAPs has remained nascent due to the lack of publicly available high-quality data (Zimmerman, 2001). With the development of content analysis techniques, a few studies have attempted to advance the understanding of MAPs by conceptualising some measures with narrative disclosure information, such as value-based management (VBM) (Firk et al., 2019; Mavropulo et al., 2021) and zero-based budgeting (ZBB) (Coyte et al.,

2020). However, conclusions of previous studies have been judged inefficient because of inadequate discussions on interdependencies amongst MAPs. Malmi and Brown (2008) and Otley (2016) explain that studies focusing on one or two dimensions of MAPs fail to provide complete and comprehensive inferences owing to a neglect of connections amongst MAPs. Therefore, this study aims to bridge the aforementioned research gaps by constructing novel and valid textual measures of MAPs from a large-scale and holistic perspective.

In this study, we first develop and validate textual measures of MAPs and then provide predictive tests of our measures by investigating the value creation function and drivers of fitted MAPs (the framework of our research is presented in Fig. 1). Distinct from previous studies centring on a certain MAP, we focus on all seven MAPs built upon the *Guides of Management Accounting Practices*¹ (short as the *Guide*) (i.e., cost, investment, budget, operations, performance, strategy, and risk). The *Guide* has been issued by the China Ministry of Finance (China MOF) since 2017 to help firms boost their competitive advantage. It provides detailed instructions on the concepts and instruments of MAPs and offers an appropriate platform for assessing the firm-level use of MAPs. Under the assumption that firms are motivated to disclose management accounting information in corporate reports to communicate with stakeholders (Firk et al., 2019), we develop firm-level textual measures of MAPs across publicly listed firms in China with a content analysis

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¹ For more information, please refer the website: http://www.mof.gov.cn/gkml/caizhengwengao/wg2016/wg201608/201612/t20161215_2483055.htm.

approach of Word2Vec. We conduct a battery of validations to verify that our measures truly reflect firms' actual choices on MAPs, including content, convergent, discriminant, persistence, reliability and mobility, and external validity. We also perform case studies of two typical reputed machine manufacturing firms to validate our novel measures of MAPs.

In predictive tests, we draw upon contingency theory, which states that no universal management accounting system is suitable for all firms (Otley, 2016), and probe the contingent fit of MAPs (*Fit*) between multi-dimension MAPs and predetermined contextual factors. Given the cluster effects amongst contextual variables, we employ life cycle stages as configurations of external environment variables and internal organisation characteristics (Miller and Friesen, 1984; Moores and Yuen, 2001). We propose theoretically ideal models of MAPs across life cycle stages and compute the cosine similarity between textual MAPs and ideal models to obtain the *Fit* index. A high similarity indicates a high level of *Fit*.

Consistent with the argument that contingent fit is a source of competitive advantage and superior performance (Chenhall, 2003), we examine its effect on firm performance to clarify the value creation function of MAPs. Using 31,999 firm-year observations across 3331 publicly listed firms spanning years from 2003 to 2019, we find that *Fit* positively influences current performance and earnings persistence. These positive effects hold with a battery of robustness checks.

Next, we conduct a differences-in-difference (DID) analysis to mitigate potential endogenous concerns that firms with excellent performance likely obtain a high contingent fit level. For this purpose, we take the event that state-owned enterprises (SOEs) are required to replace the performance indicator of return on equity (ROE) with economic value added (EVA) as an exogenous shock. The EVA policy aims to reduce overcapacity and improve firms' capital efficiency through channels of risk-taking and effective investment decisions. Given that firms in different life cycle stages are exposed to distinct MAP requirements, the effect of the EVA policy on *Fit* varies. For example, firms in the *Introduction* and *Decline* stages are expected to experience a decrease in *Fit* owing to the least requirements on risk management and investment activities. Meanwhile, contingent fit for firms in the *Growth* and *Revive* stages increases. Consistent with our assumptions, empirical results indicate that firms in the *Introduction* and *Decline* stages experience a significant increase in earnings persistence since the implementation of the EVA policy. This result provides strong evidence of the value creation of fitted MAPs.

Lastly, we explore the impact of institutional investors on *Fit*. Unlike individual shareholders, institutional investors are more professional, long-term-oriented, and motivated to exert pressure on or monitor management teams to employ appropriate MAPs (Johnson and Greening, 1999). After controlling for CEO attributes, corporate governance, and ownership concentration variables, we find that institutional investors impact *Fit* significantly and positively. Additionally, we explore the association between institutional investor heterogeneity and *Fit*, concluding that institutional investors influence corporate *Fit* through monitoring rather than the knowledge channel.

Our paper makes several contributions to the literature. Firstly, this

paper broadens the existing management accounting literature from a methodological perspective. Prior management accounting research has generally relied on such methods as surveys (Bedford et al., 2019), experiments (Dalla Via et al., 2018), and case and field studies (Dai et al., 2017; Jansen, 2011). In this study, we introduce firm-level textual measures of MAPs with the latest word embedding method of Word2Vec. Implementing such a method allows researchers to capture variations of firms' MAPs over time and detect the antecedents and consequences of MAPs in a large-scale sample. To the best of our knowledge, our study is one of the first in management accounting to utilise the Word2Vec approach and to create a model that can learn from unstructured data. The introduction of comprehensive and high-quality management accounting dictionaries to assess dynamic firm-level MAPs may inform future research that requires large-sample data on corporate MAPs.

Secondly, this research constructs a comprehensive set of MAP indices. While existing studies have focused on a single-dimension practice (e.g., balanced scorecard (BSc) (Ax and Greve, 2017); VBM (Firk et al., 2019); target costing (Yazdifar and Askarany, 2012)), our study quantifies MAPs from seven dimensions, namely, cost, investment, budget, operations, performance, strategy, and risk. The unique measures of multi-dimensional MAPs provide a dataset foundation for further studies that treat MAPs as a package and aim at unveiling the interdependencies amongst MAPs.

Thirdly, our tests of *Fit* should also enrich prior studies in the contingency theory literature. We perform relevant empirical analyses on a large-scale sample to investigate the potential determinants and consequences of fitted MAPs. Empirical results show that introducing fitted MAPs can enhance current performance and earnings persistence, and firms are motivated to employ additional fitted MAPs under considerable institutional pressure. Therefore, our study confirms *Fit*'s value creation role and helps researchers understand how institutional investors shape corporate MAPs.

The remainder of this paper is organised as follows. Section 2 briefly reviews the related research of MAPs measures. Section 3 describes the process of developing textual measures of MAPs. Section 4 presents the empirical results, testing the determinants and implications of *Fit*. Lastly, Section 5 concludes this research.

2. Research background of MAP measures

The Institute of Management Accountants (IMA) defines management accounting as “a profession that involves partnering in management decision-making, devising planning and performance management systems, and providing expertise in financial reporting and control to assist management in the formulation and implementation of an organisation's strategy” (IMA, 2008, p.48). Thus, MAPs refer to practices that facilitate the management accounting process, such as budgeting and costing (Chenhall, 2003). The literature has often quantified MAPs using the survey approach (Arnold and Artz, 2019; Bedford et al., 2019). However, surveys suffer from potential bias derived from low response rates (Hiebl and Richter, 2018) and subjective opinions from respondents (Kumar et al., 1993).

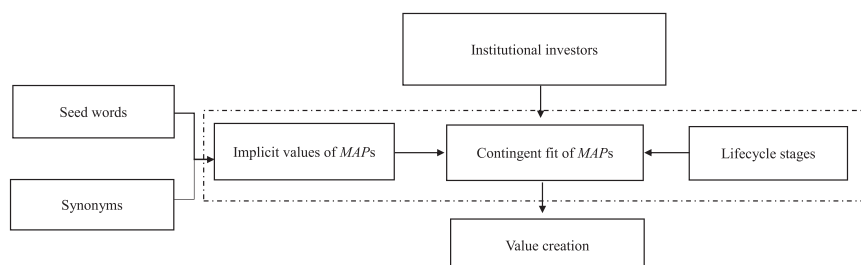


Fig. 1. Research Framework.

A few scholars have endeavoured to quantify MAPs by utilising content analysis methods, which allow them to gauge some topics of importance but are difficult to estimate (Duriau et al., 2007; Short et al., 2010). For example, Firk et al. (2019) propose a measure of VBM sophistication by collecting information on VBM elements from annual reports. Coyte et al. (2020) identify firms as ZBB adopters if firms have mentioned words of interest (i.e., zero-based budgeting, ZBB, and zero-base budgeting) in conference calls. However, current research mainly focuses on a single dimension while disregarding interdependences amongst MAPs.

This research extends existing studies and employs a novel word embedding model (i.e., Word2Vec) to measure MAPs by extracting qualitative information from annual reports. And we construct textual measures of MAPs from seven dimensions as summarised in the *Guide*. The employment of such a method facilitates building a large sample dataset of MAPs, thereby shedding new insights into future management accounting research. Moreover, the introduction of textual measures of the multi-dimensional MAPs allows us to examine MAPs from a comprehensive view.

3. Measuring MAPs using Word2Vec

We firstly apply the word embedding model Word2Vec to quantify MAPs and extensively validate our measures. We then perform predictive tests of the constructed measure of *Fit* in Section 4.

3.1. Textual measures of MAPs

Researchers in business studies have increasingly adopted content analysis methods to measure unobserved firm attributes (Duriau et al., 2007). We follow Li et al. (2021) and incorporate the deductive approach to develop textual measures of MAPs. We first develop MAP word lists and then apply them to the specific contexts of annual reports to compile narrative information.

Given the lack of well-defined finance dictionaries for MAPs, we adopt the Word2Vec approach to construct management accounting dictionaries. Compared to conventional textual analysis methods, our approach has several advantages. First, traditional textual analysis techniques treat words as independent tokens, but disregard word orders and contexts (Loughran and McDonald, 2017). In contrast, the Word2Vec method identifies synonyms by recording the number of times one specific word is proximally located with other words and expresses the words as vectors. With the premise that words with similar semantics likely appear alongside the same neighbouring words (Harris, 1954), such vectorisation effectively quantifies semantics rather than just the syntactic (Li et al., 2021). Second, different from the conventional methods that identify synonyms manually, Word2Vec is more objective and efficient because of the minimum human inputs required² (Li et al., 2021) and the use of computation technologies (LeCun et al., 2015).

Finalising MAPs dictionaries involves three steps: (1) selecting appropriate seed words; (2) expanding synonyms of seed words using the Word2Vec method; and (3) scoring MAPs with the weighted word frequencies in annual reports.

3.1.1. Seed words selection

The prerequisite for seed word selection is clarifying the scope of the target concept, which is MAPs for this study. Accordingly, we summarise and compare the categories of MAPs mentioned in the *Global Management Accounting Principles* (issued by Chartered Global Management Accountant (CGMA)), *Statements on Management Accounting* (issued by

IMA), and the *Guide* (issued by China MOF).³ While classifications from these authorities are mostly consistent, we select the *Guide* as the source of seed words for the following reasons. First, the *Guide* provides detailed instructions for each MAP in terms of concepts, contingent factors, and specific tools, thus enabling the accuracy and consistency of the selection process. Second, the dimensions embedded in the *Guide* are supported by extant management accounting academic studies. In 2017, the *Guide* included guidelines and conceptual descriptions of seven MAP dimensions (i.e., cost, investment, budget, operations, performance, strategy, and risk) along with 25 specific management accounting tools.⁴ Based on the *Guide*'s conceptual descriptions of MAPs, we identify 54 seed words (15 in cost, 7 in investment, 10 in budget, 6 in operations, 8 in performance, 5 in strategy, and 3 in risk)⁵ (details of the seed words selection are described in Appendix A in the Internet Appendix).

3.1.2. Synonyms selection

Using Word2Vec, we compute the cosine similarity amongst word vectors to measure similarities amongst words (a high degree of similarity indicates a high likelihood of words being synonyms). Afterwards, we select the top 30 synonyms for each seed word.⁶ Lastly, we review synonyms manually to ensure dictionaries are independent and exhaustive. For example, if a synonym appears in multiple dictionaries, its semantic meaning is re-evaluated to determine its final dictionary category. After cross-checking, all seven-dimensional dictionaries include 264 words in total. Words lists are presented in Table 1.

3.1.3. MAP scoring

We use the newly developed management accounting dictionaries to measure each MAP at the firm-year level, computed as the weighted total number of MAP-related words scaled by the total number of words in annual reports. Word frequency is weighted by term frequency-inverse document frequency (TF-IDF). TF indicates the MAP-related word frequencies in an annual report and IDF represents an inverse document frequency of documents containing words of interest in the corpus. The application of TF-IDF decreases the impact of frequently occurring words whilst magnifying the significance of the infrequent

³ Contents in the *Global Management Accounting Principles* include governance and risk management, strategic planning and execution, performance management and measurement, planning and forecasting, product and service delivery, and value recognition. Categories in the *Statements on Management Accounting* include business leadership and ethics, strategic cost management, planning and analysis, the future of management accounting, external reporting and disclosure management, technology enablement, operation process management and innovation, performance measurement and alignment, and risk management. Dimensions in the *Guide* include strategy, budget, cost, operations, investment, performance, and risk.

⁴ Management accounting tools described in the *Guide* consist of 2 in strategy (i.e., strategy map and value chain), 4 in budget (i.e., rolling budget, zero based budget, flexible budget, and activity based budget), 4 in cost (i.e., target costing, standard costing, variable costing, and activity based costing), 4 in operations (i.e., cost-volume-profit analysis, sensitivity analysis, marginal analysis, and benchmarking analysis), 4 in investment (i.e., discounted cash flow, project management, scenario analysis, and constrained resources optimization), 4 in performance (i.e., key performance indicator, responsibility centre, economic value added, and balanced scorecard) and 3 in risk (i.e., risk management, risk identification, and risk control).

⁵ In the robustness analysis, we exclude common seed words (i.e., collaboration, value-chain, life cycle, valuation added, project management, and scenario analysis). Results still hold (results are presented in Table D.6 in the Internet Appendix).

⁶ In the robustness analysis, we alternatively select synonyms at a similarity cut off point of 0.5. Results are the same (results are presented in Table D.7 in the Internet Appendix).

² In the process of Word2Vec implementation, limited human interventions are required to provide crucial guidance (i.e., seed words selection and final dictionary evaluation).

Table 1

Examples of Seed Words and Synonyms of MAP Dictionaries.⁷¹ This table presents examples of seed words and synonyms of MAP dictionaries.

Dimensions	Seed Words	Synonyms
Strategy	strategy management, resource allocation, collaboration, strategy map, value chain	industrial chain, industrial chain analysis, business chain, whole industry chain, supply chain, product chain, upstream and downstream, production chain, ecosphere, full value chain, corporate life cycle, customer value, business model, development mode, resource integration, develop in-depth, product line, coordinated development, business plan, quality objectives, strategic planning, organisational objectives, work plan, goal programming, operation plan, strategic development, strategic decision, brand management, cultural construction, organisational change
Budget	budget management, budgeting, operation budget, specialised budget, financial budget, comprehensive budget, rolling budget, zero-based budget, flexible budget, activity-based budget	dynamic budget, expense budget, incremental budget, cost budget, comprehensive budget management, budget control, budget assessment, planned budget, capital-plan management, budget analysis, capital budget management, quota management, budget plan, departmental budget, budget system, budgeting, expenditure budget, operating budget, company budget, cash budget, capital budget, total cost accounting, responsibility budget, target management
Cost	cost management, costing, target costing, standard costing, variable costing, activity-based cost management, life cycle, cost forecasting, cost planning, cost control, cost accounting, variable cost, activity-based costing, cost analysis, cost evaluation	straight-line amortisation method, cost accounting, replacement cost, historical cost, historical cost method, fine management, expense control, cost-reducing and efficiency increasing, process control, lean management, basic management, objective management, cost budgeting, inventory management, revenue increasing and expenses reducing, stock management, scientific management, benchmarking management, increasing income and reducing expenditures, quality cost, scheduled cost, standard cost, equivalent yield, target cost, cost indicator, project cost, dynamic cost, goal management, project cost budgeting, internal cost, budgeted cost, cost responsibility, cost of design, cost standard
Operations	operations management, valuation added, cost-volume-profit analysis, sensitivity analysis, marginal analysis, benchmarking	sensitivity analysis, quantitative analysis, impact analysis, sensitivity, rate of change, interest rate changes, price changes, regression analysis, sensitivity, factor analysis, forecasting, liquidity analysis, benchmarking management, total quality management, comprehensive quality management, amoeba, six

Table 1 (continued)

Dimensions	Seed Words	Synonyms
		sigma, lean thinking, science management, profit rate, marginal contribution, thrive in both production and marketing, operate, operation management, project management, operation, operation control, information management, business operation, operation maintenance, business management
Investment	investment management, financing management, investment and financing management, discounted cash flow, project management, scenario analysis, constrained resources optimisation	investment consulting, venture capital, industrial investment, capital management, investment centre, investment management institutions, capital investment, fund management institutions, financing investment, international investment, investment funds, venture, industry, invest, financing, crowdfunding, fund management, fundraising, capital operation management, fund plan management, capital budget management, fund control, capital operation, financing, decision management, financial management, securities management, cash management, management of investment in securities, capital operating, liquidity management, capital dispatch, contract management, organisation management, project operation, product development
Performance	performance management, key performance indicators, responsibility centre, economic value-added, balanced scorecard, equity incentive, performance appraisal, and incentive management.	performance evaluation, performance assessment, performance indicators, organisational performance, performance objectives, job performance, performance, target management, evaluation, assessment, evaluation system, assessment scheme, assessment objectives, key performance evaluation, assessment indicator, employee performance, comprehensive performance, work performance, assessment management, incentive, training management, salary management, salary incentive, pay performance, lean management, compensation system, assessment system, staff training, staff appraisal, assessment mechanism, performance incentive, equity incentive plan, motivation, employee stock ownership plan, option based payment, option incentive, stock incentive plan, employee stock ownership, stock appreciation right, share repurchase, stock repurchase, equity incentive mechanism, share buyback, rights issue, incentive mechanism, seasoned equity offering, incentive

(continued on next page)

Table 1 (continued)

Dimensions	Seed Words	Synonyms
Risk	risk management, risk identification, risk control	system, buyback, cost centre, profit centre, expense centre, responsibility centre, budget unit, departmental unit, division of labour, management responsibility system
		risk management, risk control, total risk management, risk, internal control management, operational risk, Risk monitoring, Risk prevention and control, risk prevention, liquidity management, risk management system, risk identification, liquidity risk, internal risk control, credit management, risk management department, risk management committee, system construction

ones (Henry and Leone, 2016).⁸ Lastly, we normalise MAP scores on a scale of 0–1 for the convenience of interpretation.

3.2. Validations

Given that our study is the first to measure MAPs with the content analysis method of Word2Vec, we provide the following validation tests as suggested in Short et al. (2010): (1) content validity; (2) convergent validity; (3) discriminant validity; (4) persistence, reliability and mobility; and (5) external validity.

3.2.1. Content validity

Content validity is the extent to which a measure determines the domain of the target construct (Short et al., 2010). Consistent with Short et al. (2010), we assess content validity through expert judges and contextual meaning analyses. Our two expert judges, who are familiar with management accounting terminologies, independently review the final word lists and decide whether or not they agree to include words in dictionaries. We evaluate the inter-rater reliability and obtain a Kappa value of around 0.75, which indicates a high consistency between the two judges' decisions.

For the contextual meaning analyses, we manually read the actual context of each word located in annual reports to detect how words are used in specific sentences. We list illustrative examples of sentences for selected words in each of the seven dimensions (see details in Table A.1 in the Internet Appendix). The examples show that words of interest are often used in sentences discussing the implementation of MAPs, thereby providing additional evidence for content validity.

3.2.2. Convergent validity

Given that convergent validity is of the strongest power in validating the newly established dictionaries (Gamache et al., 2020; Short et al., 2010), we regress our measures with well-proven indicators of each dimension of MAPs. We follow Li et al. (2021) and control for the variables *Size*, *ROA*, and year and industry fixed effects.

We regress the implicit cost management (*Cost*) score on the efficiency indicator of asset turnover (*Turnover*) using an ordinary least

squares (OLS) model. In management accounting studies, cost management aims to create value by improving operation efficiency (Porter, 2008). Thus, a positive association between *Cost* and *Turnover* is expected. Column (1) of Table 3 shows that *Cost* is positively related to *Turnover* (coefficient = 0.068, t-statistic = 3.130). This result implies that firms with high-weighted word frequencies of cost management in annual reports likely obtain high asset efficiency.⁹

To assess the validation of implied investment values (*Investment*), we run a logit model between *M&A* and *Investment*, where *M&A* is the indicator that takes the value of 1 if firms are involved in merger and acquisition (M&A) deals in the current year, and 0 otherwise. Based on the China MOF discussion, investment management techniques are implemented to build comprehensive evaluation mechanisms for investment projects.¹⁰ Firms are presumed to conduct additional evaluation activities when engaged in investment projects (e.g., M&A activities). Thus, observing a positive relationship between *M&A* and *Investment* is expected. Data on *M&A* is obtained from the China Stock Market and Accounting Research (CSMAR) database, which has been available since 2006. Column (2) of Table 3 shows that *M&A* is positively related to *Investment* significantly (coefficient = 1.108, t-statistic = 2.465). This result indicates that firms involved in M&A deals likely apply investment evaluation techniques.¹¹

We validate budget management (*Budget*) by examining the association between *SSC* and *Budget* using a logit model, where *SSC* takes the value of 1 if firms are identified as the 'Best Shared Service Centre' by CGMA for their excellent performance in activities of shared service centres (SSCs), and 0 otherwise. China MOF describes budget management as an instrument optimising the resource allocation process. The emergence of SSCs can revamp traditional resource top-down allocation methods into bottom-up ones via budgeting activities (Strikwerda,

Table 2

Sample Selection Process. This table represents the sample selection process for validations and empirical tests.

	#firm-year obs	# firm-year obs removed	# firms	#firms obs removed
Sample information for Table 3-Table 5				
Total scripts of annual reports spanning from 2003 to 2019	35,869		3643	
After matching with financial data for Size and ROA	33,253	2616	3425	218
Sample information for Table 8-Table 13				
Total scripts of annual reports spanning from 2003 to 2019	35,869		3643	
After deleting observations in the financial industry	35,139	730	3564	79
After matching with financial variables	31,999	3140	3331	233
After matching with CEO attributes and corporate governance variables	20,170	11,829	2626	705

⁹ In this study, we also use *Cost_Margin* (total cost/total income) as the proxy of efficiency and obtain a significant negative association between *Cost* and *Cost_Margin*, which suggests firms with higher values of cost management devote efforts to reducing costs.

¹⁰ More information is discussed in the website: http://kjs.mof.gov.cn/zhengcefabu/201710/t20171018_2727363.htm.

¹¹ In the robustness analysis, we replace *M&A* with the variable of *Capex* (value of capital expenditure) and examine the association between *Capex* and *Investment* using an OLS model. Results provide a significant and positive coefficient on *Investment*, which indicates that high investment spending firms tend to use more *Investment* techniques to obtain investment efficiency.

⁷ Each synonym provided is a translation from Chinese into English. As such, the best available translation of the synonyms has been provided, and some concepts do not have perfect translations into English.

⁸ According to the conclusion from Henry and Leone (2016), both equally weighted and TF-IDF weighted word frequency are powerful capturing qualitative information traits. Therefore, we also measure MAPs with the equally weighted method for a robustness check.

Table 3

Convergent Validity. This table represents convergent validation results for our textual measures of MAPs based on annual reports. Ordinary least squares (OLS) regressions are used when the dependent variables are *Turnover* and *Innovation*. Logit regressions are used for all other validating variables. Control variables include *Size* and *ROA*. Variables of interest are presented in bold. Model specifications include industry and year fixed effects. Robust standard errors are clustered at the industry level. Variables are defined in the Appendix. T statistics are reported in parentheses. * significant at 10%; ** significant at 5%; *** significant at 1%.

	<i>Turnover</i> (1)	<i>M&A</i> (2)	<i>SSC</i> (3)	<i>Honour</i> (4)	<i>Incentive</i> (5)	<i>Innovation</i> (6)	<i>Risk Commit</i> (7)
<i>Cost</i>	0.068*** (3.130)						
<i>Investment</i>		1.108** (2.465)					
<i>Budget</i>			1.521* (1.862)				
<i>Operations</i>				0.773*** (7.100)			
<i>Performance</i>					0.402*** (3.918)		
<i>Strategy</i>						0.516*** (4.173)	
<i>Risk</i>							2.688*** (4.392)
<i>Size</i>	0.067*** (3.210)	-0.040 (-0.425)	0.491** (2.067)	0.682*** (11.889)	0.661*** (20.836)	0.369*** (6.320)	0.663*** (5.472)
<i>ROA</i>	1.716*** (11.364)	-1.537*** (-4.171)	-1.938 (-0.397)	3.194*** (5.368)	6.108*** (10.009)	2.540*** (14.613)	3.677 (0.718)
Constants	YES	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES
Observations	33,243	12,052	10,364	29,930	33,252	14,604	21,942
Adj. R ² /Pseudo R ²	0.221	0.027	0.083	0.183	0.430	0.245	0.283

2014). Hence, we hypothesise that CGMA identified firms have superior budgeting activities compared to non-CGMA identified firms. We manually collected the SSC data, which has been available since 2015. Column (3) of Table 3 indicates a significant and positive association between *Budget* and *SSC* (coefficient = 1.521, t-statistic = 1.862), validating our textual budget management measure.

In assessing the text-based measure of operations management, we run a logit model between suggested values of operations management (*Operations*) and the achievement of product quality certifications (*Honour*), which is an indicator that takes the value of 1 if firms have achieved quality certifications or honours in the current year, and 0 otherwise. Operations management aims to maximise the value added to customers (Slack et al., 2010). Therefore, firms that accentuate operations are expected to attain quality certifications or honours. *Honour* data are collected from the Chinese Research Data Services Platform (CNRDS) database and have been available since 2006. Column (4) of Table 3 presents a significant and positive association between *Operations* and *Honour* (coefficient = 0.773, t-statistic = 7.100), which is consistent with the hypothesis that emphasis on operations management in firms is associated with optimal customer service and superb product quality.¹²

To validate the performance management measure, we test the association between performance management and executives' compensation. According to agency theory, performance management accomplishes strategic goals by aligning interests between employees and shareholders. Given the effective role of executives' compensation in the incentive scheme (Mengistae and Xu, 2004), we expect a relatively higher compensation for executives in firms that score high on *Performance*. Due to the rarity of equity compensation data (Conyon and He, 2011), we select cash compensation as a proxy for executives' compensation, which is consistent with the cash mentality of people in China (Chiu et al., 2002). Specifically, we regress implied scores of

performance management (*Performance*) on executives' cash compensation (*Incentive*) with a logit model. *Incentive* is a dummy variable for the executives' cash compensation level higher than the mean. Consistent with findings that boards adjust executives' pay for the purpose of incentive management (Buck et al., 2008), Column (5) of Table 3 shows a significant and positive association between *Incentive* and *Performance* (coefficient = 0.402, t-statistic = 3.918). This outcome validates our performance management measure.

Furthermore, we validate the textual measure of strategy management by regressing strategy management (*Strategy*) on the innovation index (*Innovation*) with an OLS model, where *Innovation* is calculated as the number of patents applied in the current year and granted in the following years. *Innovation* is collected from the WinGo database. In management accounting, strategic management activities are designed to achieve superior and sustainable performance (Porter, 2008). As an essential part of the strategy, innovation contributes substantially to achieving competitive advantage (Keupp et al., 2012). Therefore, a positive association between *Innovation* and *Strategy* is hypothesised. Column (6) of Table 3 shows that *Innovation* and *Strategy* are positively related (coefficient = 0.516, t-statistic = 4.173), thus validating our newly proposed measure of strategy management.¹³

Finally, we employ a logit regression model to examine the association between the values of risk management (*Risk*) and the formation of the risk committee (*Risk Commit*), a dummy variable that takes the value of 1 if firms establish a risk committee in the current year, and 0 otherwise. The risk committee notably improves the board's risk management function (Ishak and Nor, 2017). Thus, firms highlighting risk management activities likely establish risk committees voluntarily. Column (7) of Table 3 presents a positive association between *Risk* and *Risk Commit* (coefficient = 2.688, t-statistic = 4.392), therefore endorsing the effectiveness of the proposed risk management measure.

¹² In this study, we also use the variable of *Customer Satisfaction* (a dummy variable that equals to 1 if firms have conducted customer satisfaction surveys, and 0 otherwise) as an indicator of the products quality and obtain a significant and positive association between *Operations* and *Customer Satisfaction*.

¹³ In this study, we also regress alternative innovation index (i.e., *Exploratory innovation*) on strategy, where *Exploratory innovation* is a dummy variable that takes the value of 1 if the patent cites more than 80% of patent classification numbers that different with the existing company patent portfolio, and 0 otherwise. We obtain similar results.

Overall, the regression results confirm the high convergent validity of our measure of MAPs.

3.2.3. Discriminant validity

We follow Short et al. (2010) to assess the discriminant validity of MAPs by examining their correlation matrix. For multi-dimensional constructs of a single theoretical concept, dimensions should be distinctive, but related (Short et al., 2010). That is, correlations amongst dimensions should be significant, but not perfect. Table 4 presents the Pearson bivariate correlations. The results show that most dimensions correlate positively at the 1% significance level¹⁴ with no correlation coefficients above 0.2, suggesting that the MAP construct is multi-dimensional.

3.2.4. Persistence, reliability, and mobility

To test the persistence of each MAP value, we calculate the autocorrelations of MAPs for firms with at least 15 observations (Li et al., 2021).¹⁵ Panel A of Table 5 represents autocorrelation results between year t and years $t-1$, $t-2$, $t-3$, $t-4$, and $t-5$. The results reveal significantly positive autocorrelation coefficients between years t and $t-1$, from 0.428 to 0.616, with the highest score for *Cost*. Autocorrelations between years t and $t-2$ range from 0.275 to 0.460. By the fifth year, autocorrelations between years t and $t-5$ decrease to the range of 0.092–0.234, indicating that the persistence of MAP value diminishes gradually over time.

We employ an ANOVA estimator of two-way intra-class correlation coefficients (ICC) to test the test-retest reliability (Koo and Li, 2016). ICC denotes the contribution of firm-level factors to the total variance, ranging from 0 to 1. Perfect reliability is suggested if ICC equals 1, whereas no reliability exists if it equals 0. In Panel B of Table 5, ICCs range from 0.734 to 0.874, indicating high reliabilities for our MAP measures.

We calculate multiple mobility indices (i.e., Bartholomew and Determinant indices) to assess mobility. The Bartholomew index measures the average 'jump' of states from the starting period year $t-1$ to the destination period of year t (Bartholomew, 1973). The Bartholomew index value of 1 indicates perfect mobility, whereas the value of 0 indicates the measures are stable (i.e., immobile). The Determinant index shows opposite signs because it investigates the identity nature of two columns (Shorrocks, 1978). A high Determinant index value indicates a high immobility of the states of measures. Panel C of Table 5 shows that the Bartholomew indices for dimensions are around 0.2, and the Determinant indices are nearly 0.9. Hence, the MAP scores are stable, with low values of mobility.

3.2.5. External validity using US data

We also expand our novel measure of MAPs to the alternative setting of U.S. data to examine its external validity. We evaluate external validity by testing the generalisation ability of our findings across unique settings (Short et al., 2010). Given that the U.S. capital market research plays a crucial role in industrial and academic areas, we recalculate U.S. data-based MAP indices and validate them by regressing the indices on existing survey results. To detect the role of management practices on firm success, the U.S. Census Bureau conducted large-scale management and organisational practices survey (MOPS) in 2010 and released results from an enhanced version of the survey in 2015 (Buffington et al., 2017). This survey comprises questions on sections of monitoring, target, and incentive.¹⁶ Given most questions in the survey are related to

performance management toolkits, we test the external validity by detecting the cross-sectional relationship between our performance MAP and MOPS data. As MOPS targets manufacturing firms (their corresponding two-digit industry code in the North American industry classification system: 31–33), we regress the aggregate textual performance index (from 2011 to 2015) with the three-digit industry-level aggregate MOPS data.¹⁷ Table 6 shows a significant positive association between *Performance* and *MOPS* (coefficient = 0.322, t -statistic = 2.014), indicating the external validity of measures across multiple settings.

3.2.6. Exemplars of MAP usage: Motic vs Midea

We conduct case studies to validate MAPs further, examining two machine-manufacturing firms in China (i.e., Motic Electric and Midea). Motic Electric is one of the largest manufacturers in the Asia-Pacific region, supplying electrical insulation components, including standard products and medium- and high-voltage engineered products. Motic mainly focuses on cost leadership strategies to enhance competitive advantage and sustainability. Additionally, the firm actively implements a diversification strategy and adjusts organisation structures to respond to dynamic changes in external environments. Thus, Motic is expected to score high on *Cost* and *Investment*.

Midea is an electrical appliance manufacturer headquartered in Guangdong, China. It has become internationally renowned for its electrical products, including laundry, large cooking appliances, refrigeration appliances, etc. In 2019, Midea achieved a total sales of \$39.6 billion and ranked 312 on the Global Fortune 500 list (Fortune, 2019). In 2015, Midea was one of the 'Best Management Accounting Practice' recipients from CGMA for its excellent performance in MAPs. Midea garners a strategic competitive advantage through innovation and internal efficiency improvement channels. Therefore, balanced values for MAPs are expected for Midea.

Fig. 2 visually compares MAPs among the two firms above and other machine manufacturing firms.¹⁸ Given that the persistence of MAP implementation diminishes gradually over time, we compare the MAP values over the recent years (from 2013 to 2019). On average, machine manufacturing firms are often efficiency-oriented, thereby tending to score high in *Cost*. Fig. 2 shows that the industry average scores of *Cost*, *Strategy*, and *Operations* are 0.410, 0.445, and 0.356, respectively, which are higher than the values of remaining dimensions. Such a distribution is consistent with the industry characteristics of machine manufacturing. Consistent with our expectations, Motic outperforms the *Cost* and *Investment* dimensions, scoring above the overall manufacturing average. The high strategy score possibly results from its strategic transition from an optical microscope manufacturer to a supplier of modern medical services and from a traditional electrical business to smart energy. Midea receives balanced and high scores of multiple MAPs, particularly in *Strategy* and *Operations*, compatible with its differentiation strategy. Overall, the visualisation results effectively reflect firms' real choices of MAPs.

3.2.7. Time and industry variations of MAPs

Time-series patterns of MAPs are shown in Fig. 3, indicating that disaggregated MAPs distribute disparately over time. On average, the

¹⁴ The possible reason for the other negative results in discriminant validity tests is that MAPs operate as a package. Thus, not all MAPs are mandatory for every firm (Sandelin, 2008).

¹⁵ In this study, we test MAPs persistence with alternative thresholds of 5 or 10 observations, results are the same.

¹⁶ More information on MOPS data is available at: <https://www.census.gov/programs-surveys/mops.html>.

¹⁷ Due to data availability, we obtained the industry-level aggregate results from the management and organizational practices survey 2015, which spanning from 2011 to 2015. Therefore, in the validity test, we firstly compute the firm-level aggregate performance index from years 2011–2015.

¹⁸ The average scores of *Cost*, *Strategy*, *Performance*, *Budget*, *Risk*, *Operations*, and *Investment* for other machine manufacturing firms are 0.410, 0.445, 0.292, 0.250, 0.303, 0.356, and 0.349 respectively. The average scores of *Cost*, *Strategy*, *Performance*, *Budget*, *Risk*, *Operations*, and *Investment* for Motic are 0.994, 0.474, 0.176, 0.179, 0.312, 0.349, and 0.416 respectively. The average scores of *Cost*, *Strategy*, *Performance*, *Budget*, *Risk*, *Operations*, and *Investment* for Midea are 0.675, 0.630, 0.287, 0.364, 0.348, 0.578, and 0.516 respectively.

Table 4

Discriminant Validity. This table represents correlations between textual measures of MAPs. * significant at 10%; ** significant at 5%; *** significant at 1%.

		Cost (1)	Investment (2)	Budget (3)	Operations (4)	Performance (5)	Strategy (6)	Risk (7)
Cost	(1)	1						
Investment	(2)	0.005	1					
Budget	(3)	0.183 ***	0.027 ***	1				
Operations	(4)	0.150 ***	0.040 ***	0.084 ***	1			
Performance	(5)	0.080 ***	0.027 ***	0.056 ***	-0.019 ***	1		
Strategy	(6)	0.104 ***	0.011 **	0.102 ***	0.183 ***	-0.001	1	
Risk	(7)	0.052 ***	0.007	0.134 ***	0.137 ***	0.023 ***	0.194 ***	1

Table 5

Persistence, Reliability and Mobility. This table represents persistent, reliability, and mobility test results for our textual measures of MAPs. * significant at 10%; ** significant at 5%; *** significant at 1%.

Panel A: Persistence test										
Variable in Year <i>t</i>	Year <i>t-1</i>		Year <i>t-2</i>		Year <i>t-3</i>		Year <i>t-4</i>		Year <i>t-5</i>	
<i>Cost</i>	0.616 * **		0.460 * **		0.365 * **		0.286 * **		0.217 * **	
<i>Investment</i>	0.532 * **		0.395 * **		0.310 * **		0.252 * **		0.213 * **	
<i>Budget</i>	0.428 * **		0.275 * **		0.193 * **		0.149 * **		0.123 * **	
<i>Operations</i>	0.520 * **		0.398 * **		0.326 * **		0.271 * **		0.226 * **	
<i>Performance</i>	0.602 * **		0.448 * **		0.350 * **		0.278 * **		0.234 * **	
<i>Strategy</i>	0.475 * **		0.364 * **		0.301 * **		0.260 * **		0.226 * **	
<i>Risk</i>	0.489 * **		0.317 * **		0.221 * **		0.139 * **		0.092 * **	
Panel B: Intra-class correlation coefficients										
<i>Dimensions</i>			ICC							
<i>Cost</i>			0.874							
<i>Investment</i>			0.845							
<i>Budget</i>			0.769							
<i>Operations</i>			0.807							
<i>Performance</i>			0.853							
<i>Strategy</i>			0.807							
<i>Risk</i>			0.734							
Panel C: Mobility test										
	<i>Cost</i>	<i>Investment</i>	<i>Budget</i>	<i>Operations</i>	<i>Performance</i>	<i>Strategy</i>	<i>Risk</i>			
<i>Bartholomew</i>	0.209***	0.238***	0.222***	0.212***	0.205***	0.219***	0.202***			
<i>Determinant</i>	0.997***	0.999***	0.971***	0.994***	0.997***	0.991***	0.971***			

Table 6

External Validity. This table represents external validation results for our textual performance measure based on annual reports. Cross-sectional regression analysis is conducted. MOPS is the three-digit industry-level aggregate result from the Management and Organizational Practices Survey 2015. Control variables include aggregated *Size* and *ROA*. Variables of interest are presented in bold. Model specifications include two digits industry fixed effects. Variables are defined in Appendix. T statistics are reported in parentheses. * significant at 10%; ** significant at 5%; *** significant at 1%.

	Performance
MOPS	0.322** (2.014)
Size	0.006*** (3.021)
ROA	0.000 (0.053)
Constant	YES
Industry FE	YES
Observations	805
Adj. R ²	0.013

MAP trend is stable and steadily upward. Expectedly, *Cost* peaked in 2008 (during the financial crisis). Coincidental with the issue timeline of the supply-side structural reform policy announcement ‘to shift economy focus from scale and speed towards quality and efficiency’ (Xinhua, 2016),¹⁹ *Cost* and *Operations* have increased constantly since 2014. Conversely, the value of *Investment* reached the bottom in 2019 owing to increasing risks from increasing tariffs worldwide.

Fig. 4 shows that the distributions of disaggregate MAPs across typical industries vary along dimensions because of distinctive industrial operation emphases. Construction, information technology (IT), and machine manufacturing rank high in *Cost*, *Budget*, *Operations*, and *Strategy*. IT and culture, sports, and entertainment industries deploy additional MAPs in *Investment* due to frequent engagements in M&A deals. Wholesale and retail industries score high in *Performance* in response to challenges from uncertainties and threats from competitive markets. Finally, we include a sample from the finance industry to detect its distinct features. Fig. 4 shows that finance firms outperform in each dimension of MAPs, particularly in *Investment* and *Risk*. Evidence of such time and industry variations of MAPs provides intuitive validation of our measures.

4. Contingent fit of MAPs, firm performance, and institutional investors

To test the predictive validity of firm-level textual measures of MAPs,

¹⁹ Supply side structural reform policy in China is advanced in 2015, with goals of cutting overcapacity, reducing inventories, deleveraging, lowering costs, and strengthening weak links.

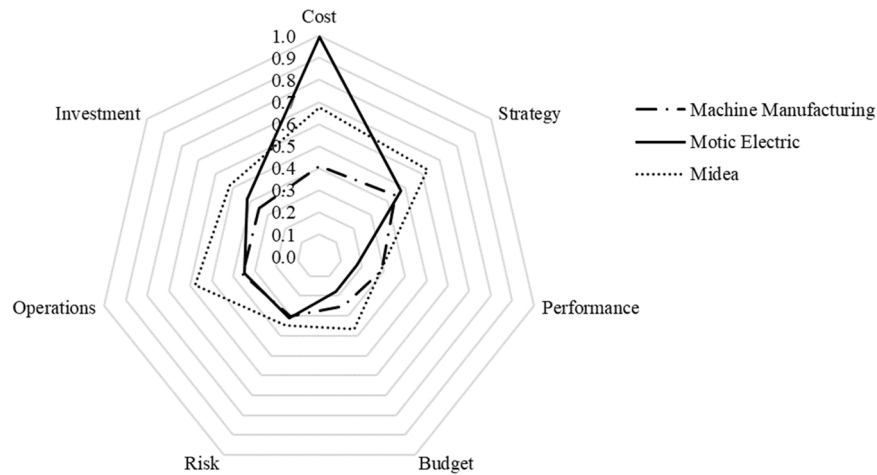


Fig. 2. Case Studies: Motic Electric (Cost leadership), Media (Differentiation), and Other Machine Manufacturing Firms.

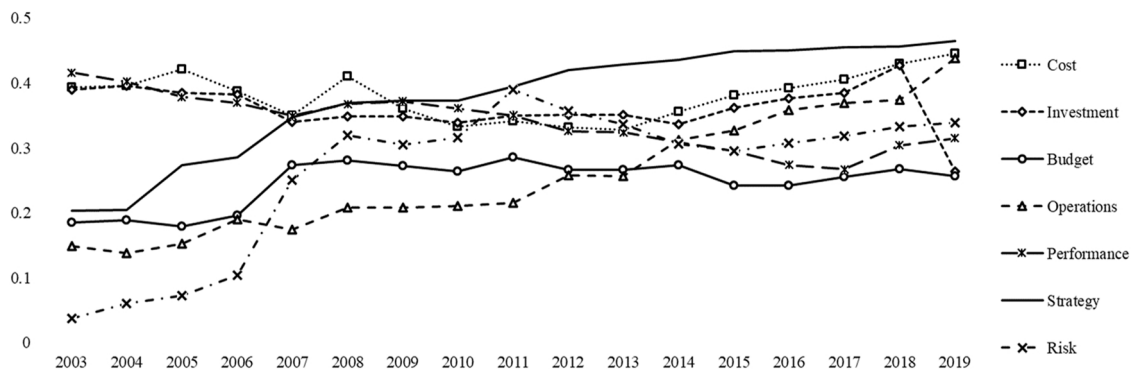


Fig. 3. Time Series Patterns of MAPs.

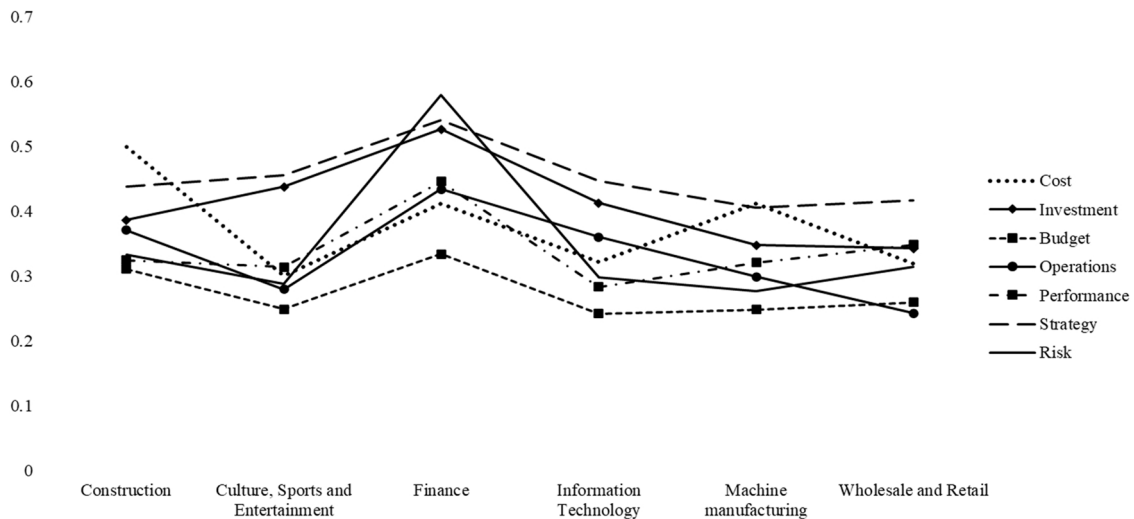


Fig. 4. Industry Variations of MAPs.

we examine the performance effect and drivers of fitted MAPs. Drawing upon contingency theory, prior research has demonstrated the importance of linking management systems to environments and organisational structures (Otley, 1980). Therefore, we develop a proxy of *Fit* by applying textual measures of MAPs and further test its impact on value creation (i.e., firm performance). In constructing *Fit*, we theoretically propose alternative ideal fit models of MAPs to contingent factors (i.e.,

life cycle stages) and then employ the cosine similarity metric to compute the similarity between ideal models and textual measures of MAPs: the more similar between the ideal and textual models, the higher level of *Fit*. Consistent with extant literature that finds firms with high fit degrees between the management accounting system and internal and external environments have high-performance levels (Chenhall, 2006), we predict a positive relationship between *Fit* and firm performance.

Additionally, we examine the role of institutional investors in determining *Fit*. Given that institutional investors are more active in voting and monitoring activities than individual investors (Huddart, 1993) and are better equipped with advanced and professional knowledge (Parrino et al., 2003), we assume a positive relationship between institutional investors and *Fit*.

4.1. Methods

4.1.1. Sample selection

To create the sample, we obtain the finance and institutional investors' shareholding data from the CSMAR database. We exclude 730 observations from the finance industry (standard industrial classification code: J) and 3140 observations with missing values of financial control variables. Our final sample for exploring the value creation of *Fit* consists of 31,999 observations across 3331 firms from 2003 to 2019. To examine the role of institutional investors on *Fit*, we exclude 11,829 observations with missing CEO attributes and ownership variables. All continuous variables are winsorised at 1% and 99% (see Appendix for variable definitions). Table 2 illustrates the sample selection process.

4.1.2. Constructing the contingent fit measure of MAPs

We obtain the *Fit* index by computing the distance between MAP indices and the theoretically proposed models with the cosine similarity method. Three additional steps are necessary: (1) constructing ideal models theoretically; (2) assigning values to MAPs in ideal models; and (3) scoring *Fit* by calculating the cosine similarity between vectors of textual MAPs and the ideal ones.²⁰

The starting point of the first step (i.e., constructing ideal models) is to identify contingent variables. Considering the interdependent relationships amongst contextual factors (Moore and Yuen, 2001), we adopt life cycle stages as configurations of contextual variables (i.e., strategy, structure, leadership style, and decision-making style) and then classify firms into five stages (i.e., *Birth*, *Growth*, *Mature*, *Revive*, and *Decline*)²¹ with the cash flow method from Dickinson (2011).²² Next, we summarise MAP heterogeneities across life cycle stages as discussed by Miller and Friesen (1984) and Moore and Yuen (2001) and propose ideal models (details of generating ideal models are presented in Appendix B in the Internet Appendix).

We next assign values (ranging from 0 to 1) to MAPs in ideal models. Practices in the ideal model with requirements of high (H), moderate (M), and low (L) levels are assigned values of 1, 0.5, and 0, respectively (details of the theoretically ideal models are presented in Table 7). Afterwards, we generate the corresponding MAP vectors for firms across stages. For example, for *Revive* firms, we can use the vector [1, 0, 0.5, 0.5, 1, 1, 0] to represent the hypothesised MAPs values in the form of [strategy, cost, performance, budget, risk, investment, operations], where demands are high for activities of strategy, risk management and

investment, moderate for performance and budget, and low for the dimensions of cost and operations.²³

Finally, we compare the angles of normalised MAP vectors between the textual-based models and proposed ideal ones to compute cosine similarity. We calculate cosine similarity by comparing the angle between two normalised vectors, and a high similarity index indicates a high value of *Fit* (see Appendix C in the Internet Appendix for details of example illustrations).

The *Fit* value for each firm-year is derived from Eq. (1) as follows:

$$Fit_{i,t} = \frac{\sum_{n=1}^7 (x_{i,n,t} * y_{i,n,t})}{\sqrt{\sum_{n=1}^7 (x_{i,n,t}^2)} * \sqrt{\sum_{n=1}^7 (y_{i,n,t}^2)}} \quad (1)$$

where $Fit_{i,t}$ indicates the value of contingent fit for firm i in year t , $x_{i,n,t}$ represents the textual value of the n^{th} MAP for firm i in year t , and $y_{i,n,t}$ demonstrates the theoretically proposed value of the n^{th} MAP for firm i in year t .

4.2. Results

4.2.1. Descriptive results

Table 8 presents the descriptive statistics. Panel A of Table 8 represents the summary statistics of the main variables. The mean (median) value of *Fit* is 0.704 (0.761), which indicates that most designs of MAPs in firms are suitable on average. In Panel B, we compare the mean values of *Fit* between firms recognised as 'Best Management Accounting Practices'²⁴ and the others. The t-test result in Panel B of Table 8 shows a significant highest value of *Fit* amongst firms rewarded, thereby indicating that our proxy can efficiently determine firms' actual situations of MAPs.

Table 9 shows the sample distribution by year and industry sectors. Panel A of Table 9 reports the sample distribution by year. *Fit* scores range from 0.558 to 0.778, steadily increasing from 2003 to 2019. The increasing trend indicates that the development of MAPs benefits substantially from recent advancements in knowledge and technology. Panel B of Table 9 provides the sample distribution across industry sectors. It shows that firms from the real estate industry operate with the lowest *Fit* score of 0.591, consistent with the real estate booms in China, where real estate firms pursue economic scale at the sacrifice of refined management. Comparatively, the highest *Fit* value for firms from competitive industries (i.e., science and technology services industry) is 0.771, above the mean value of 0.704.

Table 10 displays the correlation matrix amongst the main variables. It shows a positive correlation between firm performance (ROA) and *Fit*, which provides preliminary evidence of the positive association between *Fit* and firm performance.

4.2.2. Empirical results

Table 11 presents the empirical results examining the effect of *Fit* on firm performance with an OLS regression model. We follow the main studies in earnings (Francis and Wang, 2008; Frankel and Litov, 2009) and control for firm-level factors, including firm size (Size), leverage

²⁰ Compared to common methods measuring the contingent fit level (i.e., mediation, moderation, Euclidean distance), the cosine similarity method is advantageous in testing the fit of the entire system and less sensitive to magnitudes of points.

²¹ Dickinson (2011) forms 5 lifecycle stages (i.e., *Introduction*, *Growth*, *Mature*, *Shake-Out* (*Revive*), and *Decline*) based on the operating, investing, and financing cash flows. Cash flows in *Introduction* stage are negative in operating and investing activities, and positive in financing activities; cash flows in *Growth* stage are positive in operating and financing activities, and negative in investing activities; cash flows in *Mature* stage are negative in investing and financing activities, and positive in operating activities; cash flows in *Decline* stage are negative in operating and financing activities, and positive in investing activities; others are the ones in the *Shake-Out* (*Revive*) stage.

²² To alleviate a potential concern that cash flows may result from contingent fit levels, we also use firm age (Age) to capture lifecycles in a robustness check and obtain similar results (results are presented in Table D.5 in the Internet Appendix).

²³ In a similar vein, the vector for *Birth* firms is [1, 0, 0.5, 0.5, 1, 1, 0]; the vector [0.5, 0.5, 1, 1, 1, 1, 1] represents the ideal model for *Growth* firms; the vector for *Mature* firms is [0.5, 1, 0.5, 0.5, 0.5, 1]; the vector [0, 1, 0, 0, 0, 0] is for *Decline* firms.

²⁴ The award of 'Best Management Accounting Practices' is honoured by the CGMA for firms' excellent performance in management accounting practices. In this study, seven rewarded public firms are included in the final sample, including Midea (000333), ShenHua Holdings (600653), Haier (600690), Rui-Jie Networks (002396), Xuzhou Construction Machinery Group (000425), Chnagzhou XingYu (601799), and Fosun (600196).

Table 7

Representation of the Ideal MAPs Models. This table represents the ideal MAPs models drawing on life cycle theory.

	<i>Birth</i>	<i>Growth</i>	<i>Mature</i>	<i>Revive</i>	<i>Decline</i>
<i>Strategy</i>	High (1)	Moderate (0.5)	Moderate (0.5)	High (1)	Low (0)
<i>Cost</i>	Low (0)	Moderate (0.5)	High (1)	Low (0)	High (1)
<i>Performance</i>	Low (0)	High (1)	Moderate (0.5)	Moderate (0.5)	Low (0)
<i>Budget</i>	Low (0)	High (1)	Moderate (0.5)	Moderate (0.5)	Low (0)
<i>Risk</i>	Low (0)	High (1)	Moderate (0.5)	High (1)	Low (0)
<i>Investment</i>	Low (0)	High (1)	Moderate (0.5)	High (1)	Low (0)
<i>Operations</i>	Low (0)	High (1)	High (1)	Low (0)	Low (0)

Table 8

Descriptive Statistics. This table presents the descriptive statistics of the main variables used in the sample from 2003 to 2019. All consecutive variables are winsorised at the 1st and 99th percentiles. All variables are defined in Appendix. Panel A reports summary statistics of main variables. Panel B displays the results of the t-tests of mean values of *Fit* between firms rewarded by CGMA and the others. T statistics are reported in parentheses. * significant at 10%; ** significant at 5%; *** significant at 1%.

Panel A: Summary Statistics								
Variables	N	Mean	Min	P25	Median	P75	Max	SD
<i>ROA</i>	31,999	0.035	-0.278	0.013	0.035	0.063	0.198	0.063
<i>ROA_{t+1}</i>	28,657	0.033	-0.293	0.012	0.033	0.061	0.202	0.065
<i>ROA_{t+2}</i>	25,435	0.032	-0.302	0.011	0.031	0.059	0.207	0.066
<i>Fit</i>	31,999	0.704	0.000	0.610	0.761	0.856	0.967	0.208
<i>Size</i>	31,999	21.986	19.313	21.066	21.820	22.716	25.969	1.293
<i>Leverage</i>	31,999	0.455	0.056	0.291	0.453	0.610	1.001	0.211
<i>Growth</i>	31,999	0.207	-0.623	-0.012	0.122	0.291	3.558	0.511
<i>Age</i>	31,999	2.142	0.693	1.609	2.197	2.708	3.258	0.714
<i>Return</i>	31,999	0.206	-0.704	-0.261	0.000	0.443	3.185	0.717
<i>Volatility</i>	31,999	0.131	0.044	0.089	0.117	0.157	0.368	0.060
<i>Big4</i>	31,999	0.062	0.000	0.000	0.000	0.000	1.000	0.241
<i>Top1Shareholder</i>	31,999	35.918	8.940	23.830	33.750	46.700	75.000	15.324
<i>Duality</i>	31,999	0.223	0.000	0.000	0.000	0.000	1.000	0.416
<i>IO</i>	20,170	43.817	0.370	25.292	46.189	62.393	88.871	23.610
<i>OwnershipConcentration</i>	20,170	0.155	0.013	0.070	0.127	0.213	0.544	0.111
<i>CEOAge</i>	20,170	3.908	3.526	3.829	3.912	4.007	4.190	0.130
<i>CEOFinance</i>	20,170	0.055	0.000	0.000	0.000	0.000	1.000	0.229
<i>CEOGender</i>	20,170	0.935	0.000	1.000	1.000	1.000	1.000	0.246
<i>CEOTenure</i>	20,170	3.552	0.693	2.996	3.761	4.331	5.147	1.037
<i>Resistant</i>	20,170	4.442	0.000	0.275	1.930	6.180	29.872	6.079
<i>Sensitive</i>	20,170	1.903	0.000	0.000	0.610	2.470	16.910	3.106
<i>Indeterminant</i>	20,170	37.298	0.000	18.750	39.589	54.770	84.240	23.016
<i>Foreign</i>	20,170	0.096	0.000	0.000	0.000	0.000	2.590	0.388
<i>Domestic</i>	20,170	43.666	0.356	25.145	46.064	62.210	88.553	23.566
Panel B: Mean Values of <i>Fit</i> by Firms Type								
Mean of <i>Fit</i>		Firms Rewarded 0.857 ***		Other Firms 0.704 ***		Diff: Firms rewarded - Others 0.153 ** (2.068)		

Table 9

Sample Distribution. This table represents sample distributions. Panel A reports sample distribution by year, and Panel B reports sample distribution by industry sectors.

Panel A: Sample distribution by year				
Year	N	%	Fit	
2003	1020	3.19%	0.558	
2004	1080	3.38%	0.570	
2005	1163	3.63%	0.601	
2006	1180	3.69%	0.616	
2007	1219	3.81%	0.663	
2008	1313	4.10%	0.696	
2009	1381	4.32%	0.690	
2010	1464	4.58%	0.664	
2011	1781	5.57%	0.671	
2012	2057	6.42%	0.700	
2013	2213	6.92%	0.693	
2014	2215	6.92%	0.720	
2015	2324	7.26%	0.737	
2016	2535	7.92%	0.750	
2017	2727	8.52%	0.733	
2018	3158	9.87%	0.768	
2019	3169	9.90%	0.778	
Panel B: Sample distribution by industry sectors				
SIC code	Industry sectors	N	%	Fit
A	Agriculture	499	1.56%	0.676
B	Mining	762	2.38%	0.760
C1	Food and beverage manufacturing	2273	7.10%	0.695
C2	Apparel and paper manufacturing	5982	18.70%	0.724
C3	Machine manufacturing	11,247	35.15%	0.706
C4	Other manufacturing	567	1.77%	0.704
D	Utilities	1222	3.82%	0.762
E	Construction	827	2.58%	0.680
F	Wholesale and Retail	1878	5.87%	0.665
G	Transportation	1176	3.68%	0.743
H	Restaurants, hotel, and motel	148	0.46%	0.705
I	Information Technology	1770	5.53%	0.727
K	Real Estate	1642	5.13%	0.591
L	Leasing and Business Services	381	1.19%	0.706
M	Science and Technology Services	212	0.66%	0.771
N	Water and environment protection	351	1.10%	0.717
R	Culture, Sports and Entertainment	320	1.00%	0.750
Others		129	0.40%	0.736

ratio (*Leverage*), income growth (*Growth*), age (*Age*), stock return (*Return*), stock volatility (*Volatility*), auditing quality (*Big4*) and corporate governance variables (*Top1Shareholder*, *Duality*). Both Industry and Year fixed effects are controlled. Standard errors are clustered at the firm level.

Column (1) of Table 11 shows a positive association between *Fit* and current year performance (*ROA*) at the 1% significance level (coefficient = 0.026; t-statistic = 13.478). It illustrates that fitted *MAPs* are significantly related to firms' earnings performance. Columns (2) and (3) demonstrate the impact of *Fit* on earnings persistence. The interaction of *Fit*ROA* is the variable of interest. Results show positive associations between *Fit* and earnings in the following two consecutive years

(Column (2): coefficient = 0.369, t-statistic = 7.627; Column (3): coefficient = 0.411, t-statistic = 7.869). This result provides empirical evidence of the enduring impact of appropriate employment of *MAPs* with specified contextual environmental factors.

Furthermore, we conduct a battery of robustness checks: (1) controlling for firm fixed effects; (2) replacing the TF-IDF weighted *Fit* ratio as the equally weighted *Fit* w ratio; (3) replacing the dependent variable of *ROA* as *ROE*; (4) excluding 'large' firms; (5) using firm age as an alternative measure of life cycle stages; (6) replacing dictionaries with those excluding common words; (7) replacing dictionaries with those selecting synonyms at a similarity cut off point of 0.5; and (8) including additional control variables of CEO attributes and ownership

Table 11

Contingent Fit of *MAPs* and Firm Performance. This table reports OLS results to examine the effect of *Fit* on *ROA*. The dependent variable *ROA* is the earnings ratio, measured as net income divided by lagged total assets. The independent variable of interest is the contingent fit of the weighted textual measures of *MAPs*. Weight is TF-IDF. Control variables include *Size*, *Leverage*, *Growth*, *Age*, *Return*, *Volatility*, *Big4*, *Top1Shareholder*, and *Duality*. Variables of interest are presented in bold. Model specifications include industry and year fixed effects. Robust standard errors are clustered at the firm level. T statistics are reported in parentheses. * significant at 10%; ** significant at 5%; *** significant at 1%.

	ROA _t (1)	ROA _{t+1} (2)	ROA _{t+2} (3)
<i>Fit_t</i>	0.026*** (13.478)	0.012*** (4.812)	0.006** (2.117)
<i>ROA_t</i>		0.170*** (4.881)	0.039 (1.045)
<i>Fit_t*ROA_t</i>		0.369*** (7.627)	0.411*** (7.869)
<i>Size</i>	0.010*** (17.533)	0.000 (0.231)	-0.001 (-1.615)
<i>Leverage</i>	-0.135*** (-39.335)	-0.031*** (-10.503)	-0.028*** (-8.100)
<i>Growth</i>	0.021*** (26.396)	0.004*** (4.472)	-0.001 (-1.231)
<i>Age</i>	-0.007*** (-8.871)	-0.002*** (-3.787)	-0.000 (-0.334)
<i>Return</i>	0.025*** (32.145)	0.014*** (15.923)	0.010*** (10.858)
<i>Volatility</i>	-0.157*** (-17.811)	-0.075*** (-7.907)	-0.077*** (-6.799)
<i>Big4</i>	-0.001 (-0.385)	0.007*** (4.156)	0.008*** (3.800)
<i>Top1shareholder</i>	0.000*** (7.691)	0.000*** (8.845)	0.000*** (8.228)
<i>Duality</i>	-0.002 (-1.375)	-0.002** (-2.104)	-0.003*** (-2.725)
Constant	YES	YES	YES
Industry FE	YES	YES	YES
Year FE	YES	YES	YES
Cluster	Firm	Firm	Firm
Observations	31,999	28,657	25,435
Adj. R ²	0.299	0.269	0.157

Table 10

Correlation Matrix. This table presents the Pearson correlation matrix of key variables in the main regression analyses. All consecutive variables are winsorised at the 1st and 99th percentiles. * significant at 10%; ** significant at 5%; *** significant at 1%.

	ROA	Fit	Size	Leverage	Growth	Percent	CEO	Age	Return	Volatility
ROA	1									
Fit	0.175 ***	1								
Size	0.072 ***	0.190 ***	1							
Leverage	-0.376 ***	-0.134 ***	0.344 ***	1						
Growth	0.193 ***	-0.005	0.043 ***	0.048 ***	1					
Top1Shareholder	0.113 ***	0.020 ***	0.188 ***	0.039 ***	0.033 ***	1				
Duality	0.017 ***	0.007	-0.109 ***	-0.112 ***	0.024 ***	-0.069 ***	1			
Age	-0.145 ***	-0.013 **	0.329 ***	0.319 ***	-0.035 ***	-0.087 ***	-0.171 ***	1		
Return	0.115 ***	-0.005	-0.058 ***	0.037 ***	0.095 ***	-0.006	-0.034 ***	0.014 **	1	
Volatility	-0.075 ***	-0.040 ***	-0.173 ***	0.034 ***	0.056 ***	-0.067 ***	0.037 ***	-0.040 ***	0.457 ***	1
Big4	0.055 ***	0.079 ***	0.338 ***	0.070 ***	-0.011 *	0.136 ***	-0.058 ***	0.059 ***	-0.006	0.071 ***

concentration. The findings confirm evidence from the main results. The Internet Appendix presents detailed results in Tables D.1–D.8.

A potential endogenous problem with our results on value creation is that *Fit* is likely high when firms obtain high performance. To mitigate such a concern, we conduct a DID analysis to compare the changes of *ROA* in observations with an increased *Fit* from years before to after the introduction of the policy, replacing ROE with EVA as one of the performance indicators for SOEs since 2010, relative to the changes of *ROA* in observations with decreased *Fit*. Given that this policy aims to improve the capital efficiency of SOEs, we reduce the sample to SOEs from 2006 to 2014. Furthermore, we drop observations in the event year 2010. Initially, we empirically identify firms exposed to the regulation differently. Because of variations of MAPs required, *Fit* levels change dissimilarly across life cycle stages. For example, given that the focus of the regulation is on risk-taking and investing behaviours, firms in stages with minimum requirements of risk management and investment activities (i.e., *Introduction* and *Decline*) are supposed to suffer from a decrease in *Fit*. By contrast, firms in the *Growth* and *Revive* stages benefit from an increase in *Fit*.²⁵ Given the lagged effect of the regulation, we examine its effect on firms' earnings persistence.

Table 12 shows the DID result from the OLS model, in which the variable *Post*Treated*ROA* is of interest (*Post* equals 1 for years after 2010, and 0 otherwise; *Treated* takes the value of 1 if firms are in the *Growth* and *Revive* stages, and 0 otherwise). The results in Table 12 present a positive coefficient on the variable *Post*Treated*ROA* (coefficient = 0.110, t-statistic = 1.826), showing that firms at the life cycle stages of *Introduction* and *Decline* (*Growth* and *Revive*) endure a(n) decrease (increase) of earnings persistence. This outcome indicates that firms benefit from the increase of *Fit*, which results from the regulation replacing ROE with EVA, and thus further supports existing results.²⁶

Table 13 empirically tests the impact of institutional investors (*IO*) on *Fit* with an OLS regression model. Given the role of CEOs and corporate governance systems in firms' operational processes, we control for CEO attribute variables of age (*CEOAge*), finance background (*CEOFinance*), gender (*CEOGender*), tenure (*CEOTenure*), corporate governance variables (*Top1Shareholder*, *Duality*), and firms' ownership concentration (*OwnershipConcentration*). Industry and Year fixed effects are controlled for. Standard errors are clustered at the firm level. Column (1) of Table 13 shows a significant and positive association between *IO* and *Fit* (coefficient = 0.001, t-statistic = 5.429), supporting our argument that institutional investors can contribute to enhancing corporate *Fit* levels. Consistent with findings in previous studies (Bhimi, 2009; Davila and Foster, 2005; Ma et al., 2010), we also observe significant positive coefficients on corporate governance, ownership concentration, CEO age, and tenure. Moreover, the significant negative coefficient of *CEOFinance* indicates that financial expert CEOs are relatively less skilled in nonfinancial decision-making processes (e.g., innovation, advertising, human resource management) (Custódio and Metzger, 2014) owing to limited capabilities in aligning strategies with resources (Li and Patel, 2019).

Additionally, we investigate the relationship between institutional investor heterogeneity and *Fit* from potential channels of monitoring and knowledge. From the perspective of the monitoring mechanism,

Table 12

DID Analysis. This table reports DID test results to examine the effect of *Fit* on *ROA* from 2006 to 2014 (the event year of 2010 was dropped). The independent variable of interest is the interaction variable of *Post*Treated*ROA_t*. Where *Post* denotes time, equal to 1 if years are after 2010, and 0 otherwise. *Treated* denotes firms, equal to 1 if firms are at stages of *Growth*, *Revive* in year 2010, and 0 otherwise (we drop *Mature* firm due to ambiguous predictions). Control variables are the same as the ones in the main results. Model specifications include industry and year fixed effects. Robust standard errors are clustered at the firm level. T statistics are reported in parentheses. * significant at 10%; ** significant at 5%; *** significant at 1%.

	<i>ROA_{t+1}</i>
<i>ROA_t</i>	0.293*** (7.039)
<i>Treated</i>	0.007** (2.162)
<i>Post*Treated</i>	-0.006 (-1.403)
<i>Post*Treated*ROA_t</i>	0.110* (1.826)
Controls	YES
Constant	YES
Industry FE	YES
Year FE	YES
Cluster	Firm
Observations	4328
Adj. R ²	0.189

studies of the literature have found that institutional investors are unequally motivated to monitor and discipline corporate management (Almazan et al., 2005; Chen et al., 2007). We follow Brickley et al. (1988) and classify institutions into three groups (i.e., pressure-resistant, pressure-sensitive, and pressure-indeterminate institutions) to test the monitoring channel. In line with the statement that pressure-resistant institutions, institutions lacking business relationships with their portfolio firms, are more successful in monitoring than the remaining institutions, we expect that pressure-resistant institutions drive the positive effect of institutional investors on *Fit* under the monitoring channel. From the knowledge channel perspective, institutional investors' effect on firm decisions varies on the ability to provide professional knowledge (Neupane et al., 2016). We alternatively sort institutions into foreign and domestic ones. Given that domestic institutions have more local knowledge (Neupane et al., 2016), especially on environments determining MAPs, we assume a more significantly positive effect of domestic institutional investors on *Fit*.

Column (2) and (3) in Table 13 report the association results of institutional investor heterogeneity and *Fit*. Column (2) results only suggest significant positive effects of pressure-resistant and indeterminate institutions on *Fit*, supporting the monitoring channel. For the knowledge channel results, Column (3) presents significant positive coefficients on both variables of foreign and domestic institutions. Therefore, the positive effect of institutional investors on *Fit* is likely to be realised through monitoring rather than a knowledge channel. To alleviate the potential effect of government policy on the choices of MAPs, we further reduce the sample to the voluntary one, excluding SOEs after 2010.²⁷ Columns (4)–(6) of Table 13 report consistent results from the voluntary sample.

²⁵ As firms at the *Mature* stage require risk management and investment activities at medium levels, it is hard to predict the effect of EVA policy on *Fit*. Thus, we delete observations from the *Mature* stage to ensure results robustness.

²⁶ Given the considerable number of firms at the *Mature* stage, we also replicate DID analyses with alternative samples including *Mature* firms into treated or control groups. Results indicate a significant and positive coefficient on the interactive variable of *Post*Treated*ROA_t* (coefficient = 0.151, t-statistic = 3.523) when *Mature* firms are sorted into the control group but an insignificant one (coefficient = 0.021, t-statistic = 0.367) when *Mature* firms are defined as the treated ones, which suggests a potential negative effect of EVA policy on *Mature* firms due to their conservative attributes and thus support our results.

²⁷ In 2010, the State-Owned Assets Supervision and Administration Commission (SASAC) mandatorily require replacing Return on Equity (ROE) with Economic Value Added (EVA) as performance indicators of SOEs.

Table 13

Determinants of Contingent Fit of MAPs This table reports OLS results to examine the effect of *IO* on *Fit*. Columns (1)–(3) ((2)–(4)) are derived from the full (voluntary) samples. In the voluntary sample, we exclude SOEs since 2010. The dependent variable *Fit* is the contingent fit of the weighted textual measures of MAPs. Weight is TF-IDF. Control variables include *Top1Shareholder*, *Duality*, *CEOAge*, *CEOFinance*, *CEOGender*, *CEOTenure*, and *OwnershipConcentration*. Variables of interest are presented in bold. Model specifications include industry and year fixed effects. Robust standard errors are clustered at the firm level. Variables are defined in Appendix. T-statistics are reported in parentheses. * significant at 10%; ** significant at 5%; *** significant at 1%.

	<i>Fit</i>					
	Full sample			Voluntary sample		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>IO</i>	0.001*** (6.984)			0.001*** (4.998)		
<i>Resistant</i>		0.003*** (11.810)			0.003*** (10.682)	
<i>Sensitive</i>		0.000 (0.020)			-0.000 (−0.484)	
<i>Indeterminant</i>		0.000*** (3.739)			0.000*** (2.650)	
<i>Foreign</i>			0.021*** (5.461)			0.025*** (5.395)
<i>Domestic</i>			0.001*** (6.568)			0.000*** (4.581)
Controls	YES	YES	YES	YES	YES	YES
Constant	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES
Year fixed FE	YES	YES	YES	YES	YES	YES
Cluster	Firm	Firm	Firm	Firm	Firm	Firm
Observations	20,170	20,170	20,170	13,249	13,249	13,249
Adj. R ²	0.070	0.076	0.071	0.062	0.070	0.064

5. Conclusion

MAPs act as a pivotal component of value creation in organisations. With escalating environmental uncertainties, improving operations through effective MAPs is increasingly crucial for firms to survive. However, properly measuring MAPs remains a major concern for investors, regulators, and policy makers. This study proposes comprehensive and large sample scale-based measures of MAPs with a word embedding model to extract MAP-related words in annual reports and score MAPs as weighted word frequencies.

Furthermore, we validate MAP measures from perspectives of content, convergent, discriminant, persistence, reliability and mobility, and external validity. We also perform case studies and represent time and industry variations of MAP distributions, thereby validating our measures intuitively. Drawing on contingency theory, we construct a contingent fit index of MAPs and examine the determinants and consequences of *Fit*. Empirical results suggest a positive association between *Fit* and firm performance and document a positive impact of institutional investors on *Fit*.

Our textual measures of MAPs enable us to directly evaluate firm-level MAPs and are valuable for future research opportunities requiring dynamic measures of corporate MAPs. Additionally, novel multi-dimensional MAPs measures enable management accounting researchers to comprehensively conduct empirical tests of MAPs implications concerning the interdependences amongst MAPs. Finally, we add further empirical evidence of *Fit*, enriching the contingency theory literature and presenting potential research opportunities with our newly developed measures.

Apart from the contributions above, some limitations must be discussed as well. First, our textual measure of MAPs is developed in the Chinese setting. Although we provide a clear blueprint for developing similar measures in appendices and expanding performance dimension

data to the U.S. context, further extending this firm-level approach in the setting studies of other countries (e.g., the U.S. and European countries) can provide substantial contributions. Second, in developing the *Fit* index, we theoretically propose ‘ideal’ models of MAPs for life cycle stages and assign values from 0 to 1 accordingly. We admit potential inaccuracies with such an approach and therefore call for alternative approaches in further research to refine the method in determining *Fit*.

Several future research avenues remain to be examined. Future studies can follow our approach and use alternative data sources (e.g., conference calls, management discussion and analysis) to provide complementary evidence or refine our textual measures of MAPs. Moreover, additional factors behind contingent fit variations can be further explored to enrich current findings. For example, given that top team management (TMT) plays a crucial role in influencing the implementation of MAPs, the effects of TMT characteristics on contingent fit should be investigated. Further investigations on potential outcomes from MAPs may help advance our understanding of MAPs. For example, studies on operational outcomes (e.g., operation leanness) and capital market reactions could provide future insights into the use and impact of MAPs.

This study also provides practical implications. First, annual reports appear to be an effective channel for communicating the adoption of MAPs with stakeholders. Additionally, appropriate MAPs are found to play a significant role in value creation and should be included when analysing firm valuations. Finally, different stages of firm development require different implementations of MAPs, or rather, dynamic implementations of MAPs across the life cycle stages. Thus, effective decisions on MAP adoption should be linked with contingent factors.

Data availability

Data will be made available on request.

Appendix: Variable Definitions

Variable	Definitions
Management accounting variables	
<i>Cost</i>	Weighted word frequency of Cost-related words in annual reports
<i>Investment</i>	Weighted word frequency of Investment-related words in annual reports
<i>Budget</i>	Weighted word frequency of Budget-related words in annual reports
<i>Operations</i>	Weighted word frequency of Operations-related words in annual reports
<i>Performance</i>	Weighted word frequency of Performance-related words in annual reports
<i>Strategy</i>	Weighted word frequency of Strategy-related words in annual reports
<i>Risk</i>	Weighted word frequency of Risk-related words in annual reports
<i>Fit</i>	Cosine similarity between vectors of the weighted text-based MAPs and the ones in theoretically proposed models (see Appendix B in the Internet Appendix for details). The weight is TF-IDF
Validating variables	
<i>Turnover</i>	(Total income/lagged total assets)* 100%
<i>M&A</i>	Dummy variable, which takes the value of 1 if firm <i>i</i> is engaged in the merger and acquisitions deals in year <i>t</i> , and 0 otherwise
<i>SSC</i>	Dummy variable, which takes the value of 1 if firm <i>i</i> is rewarded as the 'Best Shared Service Centre' by CGMA in year <i>t</i> , and 0 otherwise
<i>Honour</i>	Dummy variable, which takes the value of 1 if firm <i>i</i> achieved certification and honours in product quality in year <i>t</i> , and 0 otherwise
<i>Incentive</i>	Dummy variable, which takes the value of 1 of the total compensation values of the top 3 executives are larger than the mean value, and 0 otherwise
<i>Risk Commit</i>	Dummy variable, which takes the value of 1 if firms have established the risk committee, and 0 otherwise
<i>Innovation</i>	Natural log of one plus the number of patents applied at the current year and granted in the following years
<i>MOPS</i>	Three digits industry-level aggregate results from Management and Organisational practices survey 2015 on US manufacturing firms (from 2011–2015). For more details, see https://www.census.gov/programs-surveys/mops.html
Implication variables	
<i>ROA</i>	(Net income/ total asset at the ending of the year) * 100%
<i>IO</i>	Shareholding percent of institutional shareholders
<i>Resistant</i>	Total shareholding percent of institutional shareholders from institutions of fund, qualified foreign institutional investors, and security fund
<i>Sensitive</i>	Total shareholding percent of institutional shareholders from institutions of broker, insurance, entrust, finance, and bank
<i>Indeterminant</i>	Total shareholding percent of institutional shareholders from other institutions that are not included in the other two groups
<i>Foreign</i>	Total shareholding percent of institutional shareholders from institutions of qualified foreign institutional investor
<i>Domestic</i>	Total shareholding percent of institutional shareholders from domestic institutions
Difference in Difference (DID) variables	
<i>Post</i>	Dummy variable, which takes the value of 1 for observations in years after 2010, and 0 otherwise
<i>Treated</i>	Dummy variable, which takes the value of 1 if observations are at the stage of <i>Growth</i> , <i>Revive</i> in year 2010, and 0 otherwise
Control variables	
<i>Size</i>	Log total assets
<i>Leverage</i>	(Total liabilities/total assets) * 100%
<i>Growth</i>	(Total income _{<i>t</i>} -total income _{<i>t-1</i>}) / total income _{<i>t-1</i>} * 100%
<i>Age</i>	Natural log of one plus the firm age, where the age is based on the number of years the firm listed in Shanghai and Shenzhen Stock Exchanges
<i>Return</i>	Annual return on individual shares with cash dividends reinvested for firm <i>i</i> in year <i>t</i>
<i>Volatility</i>	Standard deviation of monthly stock returns for firm <i>i</i> in year <i>t</i>
<i>Big4</i>	Dummy variable, which takes the value of 1 if firm <i>i</i> is audited by one of Big 4 auditing firms in year <i>t</i> , and 0 otherwise
<i>Top1Shareholder</i>	Shareholding percent of the biggest shareholder
<i>Duality</i>	Dummy variable, which takes the value of 1 if CEO and chairman are the same people, and 0 otherwise
<i>CEOAge</i>	Natural log of one plus the CEO age
<i>CEOFinance</i>	Dummy variable, which takes the value of 1 if CEO in firm <i>i</i> has past experience in financial industries (banking, investment, insurance, security), and 0 otherwise
<i>CEOGender</i>	Dummy variable, which takes the value of 1 if CEO in firm <i>i</i> is a male, and 0 otherwise
<i>CEOTenure</i>	Natural log of one plus the number of years the current CEO is in the CEO office of firm <i>i</i>
<i>OwnershipConcentration</i>	Sum of squared share percent of the top three institutional investors for firm <i>i</i> in year <i>t</i>

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.mar.2022.100818](https://doi.org/10.1016/j.mar.2022.100818).

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